Environmental Data Science: Homework #9
Boosting
Due: Before Class on 11/04/26

Boosted regression in python – NOAA Buoys

We'll be using data from a NOAA buoy off the coast of California, same as last week. A link to the data descriptions can be found here:
www.ndbc.noaa.gov/station_page.php?station=46222

Data are posted on brightspace. [The file is whitespace delimited, so pd.read.csv will not work by default.]

Using these data, train models to predict wave height as a function of dominant wave period, average wave period, the direction from which the waves at the dominant period are coming, and sea surface temperature. Use the XGBoost package for boosting.

1. Fit a boosted regression tree with 25 boosting iterations. Compare with a baseline linear model of your choice. Which model performs best? Report performance on the training, validation, and test dataset.
2. What cost function are you using for each model?
3. What (if any) hyperparameters did you optimize, and why?
4. Train a boosted regression tree on this prediction task with 2, 10, 25, 50, 100, 250, 500, & 1,000 boosting iterations. Plot the training and validation error as a function of number of iterations.
5. Do you have evidence of overfitting to the training dataset with 1,000 boosting iterations?
6. Retrain the boosted regression tree using early stopping with a tolerance of 10. How many boosting iterations were optimal?

Final Project Proposal

Write a 1.5-page proposal for your final project. Using single-spaced size 12 font. Full credit for thoughtful completion – the goal here is to make sure you're all on track. Include the following:

1. Title
2. Background/Introduction
3. Data Description
    a. Include proof that you can load the data in python!

4. A plot of some of your data
5. Machine Learning Methodology
    a. Include method description and appropriate baselines.
6. Scientific Innovation of the Approach
    a. What might you learn from this? How could this be valuable?
7. References